**Table 1:** Executed simulation studies of QEM Explorer 1.8.0

| Study | Status | Tasks | Circuits | Budgets | Candidates | $r$/half | $K$/half | Seeds |
|---|---|---|---|---|---|---|---|---|
| Development (dev18) | development | 88 | 44 | 10 | 17 | 60 | 6 | 2 |
| Confirmation (conf18) | confirmatory | 58 | 29 | 10 | 17 | 60 | 6 | 2 |
| Regime study (regimes18) | exploratory | 21 | 3 | 10 | 17 | 60 | 6 | 2 |
| Noise composition (noisecomp18) | exploratory | 18 | 3 | 10 | 17 | 60 | 6 | 2 |
| Root audit (4 roots) | exploratory | 8×4 | 8 | 3 | 17 | 40 | 6 | 2 |

All results are fake-device simulations, not hardware measurements. Circuits are independent logical (base) circuits; a task is a circuit with an input, observable, layout and device. The controlled studies reuse circuits of earlier releases' exploration sets and must not be added to obtain a total independent sample size. Development: out-of-fold, the configuration frozen once on it; confirmation: new circuits, evaluated once after the freeze. Seeds: seed scheme (2 = namespaced preparation streams; finding N1). K = fresh training designs per label half for the learning candidates.

**Table 2:** Candidates of portfolio-1.8 and their shot ledgers

| Candidate | Family | Validation | Feasible $10^3$ | Feasible $10^5$ | Feasible $10^7$ | Cal. | Train. | Target |
|---|---|---|---|---|---|---|---|---|
| RAW | raw | direct | 88/88 | 88/88 | 88/88 | 0 % | 0 % | 100 % |
| RO | raw | direct | 88/88 | 88/88 | 88/88 | 13 % | 0 % | 87 % |
| ZNE-L | zne | direct | 88/88 | 88/88 | 88/88 | 7 % | 0 % | 93 % |
| ZNE-R | zne | direct | 88/88 | 88/88 | 88/88 | 4 % | 0 % | 96 % |
| ZNE-E | zne | both | 88/88 | 88/88 | 88/88 | 7 % | 0 % | 93 % |
| PEC | pec | direct | 88/88 | 88/88 | 88/88 | 11 % | 0 % | 89 % |
| CDR | learning | both | 0/88 | 82/88 | 82/88 | 0 % | 25 % | 75 % |
| NIL | nil | both | 0/88 | 70/88 | 70/88 | 4 % | 50 % | 47 % |
| RO-TREX * | raw | reference | 88/88 | 88/88 | 88/88 | 6 % | 0 % | 94 % |
| RO-TW * | raw | direct | 88/88 | 88/88 | 88/88 | 13 % | 0 % | 87 % |
| RO-DD * | raw | direct | 88/88 | 88/88 | 88/88 | 13 % | 0 % | 87 % |
| ZNE-L-TW * | zne | direct | 88/88 | 88/88 | 88/88 | 7 % | 0 % | 93 % |
| ZNE-E-B * | zne | reference | 88/88 | 88/88 | 88/88 | 7 % | 0 % | 93 % |
| PEC-TW * | pec | direct | 88/88 | 88/88 | 88/88 | 11 % | 0 % | 89 % |
| FR * | learning | reference | 0/88 | 81/88 | 81/88 | 0 % | 25 % | 75 % |
| SV * | symmetry | reference | 16/88 | 16/88 | 16/88 | 11 % | 0 % | 89 % |
| ZNE-L∘SV * | chain | reference | 16/88 | 16/88 | 16/88 | 5 % | 0 % | 95 % |

Feasible: development tasks on which the candidate's allocation is feasible at the total budget B (feasible/all). Ledger: median share of B over feasible development tasks at B = $10^5$ spent on calibration, training and the target circuits (rejected post-selection shots count as target). Validation: direct = agrees with direct shot-by-shot execution; reference = agrees with an independently written shot-level reference; both = direct+reference. New in 1.8.0: marked *.

**Table 3:** Once-only confirmation on 29 new logical circuits (580 rows)

| Policy | Mean $\ell$ [95%] | $\Delta\ell$ vs SBS-B [95%] | Mean MSE | $R > 10R_{\mathrm{RAW}}$ | $q_{95}$ risk |
|---|---|---|---|---|---|
| RAW | $-3.178$ $[-3.470, -2.837]$ | $+1.175$ $[+0.968, +1.411]$ | $8.04 \times 10^{-3}$ | 0.0 % | $8.10 \times 10^{-2}$ |
| SBS | $-4.144$ $[-4.387, -3.865]$ | $+0.208$ $[+0.067, +0.380]$ | $1.35 \times 10^{-3}$ | 0.0 % | $6.74 \times 10^{-3}$ |
| SBS-B | $-4.352$ $[-4.493, -4.175]$ | $+0.000$ $[+0.000, +0.000]$ | $8.64 \times 10^{-4}$ | 0.0 % | $4.58 \times 10^{-3}$ |
| SIZE-B | $-4.366$ $[-4.526, -4.204]$ | $-0.013$ $[-0.130, +0.103]$ | $7.78 \times 10^{-4}$ | 0.0 % | $3.95 \times 10^{-3}$ |
| HGB (D18+S), frozen | $-4.511$ $[-4.644, -4.366]$ | $-0.159$ $[-0.253, -0.086]$ | $6.09 \times 10^{-4}$ | 0.0 % | $2.55 \times 10^{-3}$ |
| HGB (D) | $-4.444$ $[-4.597, -4.280]$ | $-0.092$ $[-0.204, +0.010]$ | $9.07 \times 10^{-4}$ | 0.0 % | $2.55 \times 10^{-3}$ |
| Ridge (D) | $-4.482$ $[-4.622, -4.329]$ | $-0.130$ $[-0.226, -0.037]$ | $6.56 \times 10^{-4}$ | 0.0 % | $2.55 \times 10^{-3}$ |
| Structured (D) | $-4.367$ $[-4.499, -4.211]$ | $-0.014$ $[-0.125, +0.085]$ | $1.00 \times 10^{-3}$ | 1.7 % | $5.31 \times 10^{-3}$ |
| Selective HGB (D18+S) | $-4.511$ $[-4.644, -4.366]$ | $-0.159$ $[-0.253, -0.086]$ | $6.09 \times 10^{-4}$ | 0.0 % | $2.55 \times 10^{-3}$ |
| Full policy, default bundle | $-4.436$ $[-4.578, -4.283]$ | $-0.084$ $[-0.179, -0.002]$ | $5.70 \times 10^{-4}$ | 0.0 % | $2.55 \times 10^{-3}$ |
| Full policy, simulator bundle | $-4.496$ $[-4.628, -4.354]$ | $-0.144$ $[-0.231, -0.076]$ | $5.38 \times 10^{-4}$ | 0.0 % | $2.55 \times 10^{-3}$ |
| Empirical reference | $-4.653$ $[-4.751, -4.540]$ | $-0.301$ $[-0.406, -0.210]$ | $3.10 \times 10^{-4}$ | 0.0 % | $1.32 \times 10^{-3}$ |

All budgets are averaged within a base circuit, then circuits receive equal weight. The primary objective is l = log10(MSE + $10^{-6}$); $\Delta\ell$ = l(policy) - l(SBS-B), paired percentile bootstrap over base circuits (1000 resamples). A difference $\Delta\ell$ changes the geometric mean of (MSE + $10^{-6}$) by the factor $10^{\Delta\ell}$; it is not a percentage change of the MSE. Mean MSE: arithmetic, circuit-weighted. Severe (risk > 10 x RAW's risk): declared severe degradation (share of rows); q95: 95 % quantile of the chosen risk. D18 = deployable features of 1.8.0; D = the deployable features of 1.7.0 (without the 1.8.0 noise-composition features), kept as frozen comparators; +S = with simulator-only features. Model rows are model components (the model's best label-feasible candidate); the full-policy rows call the deployment API itself (guard, out-of-distribution checks, preflight, margin rule, fallback) and are the application's result.

**Table 4:** Development performance, grouped five-fold out-of-fold (44 circuits, 880 rows)

| Policy | Mean $\ell$ [95%] | $\Delta\ell$ vs SBS-B [95%] | Mean MSE | $R > 10R_{\mathrm{RAW}}$ | $q_{95}$ risk |
|---|---|---|---|---|---|
| RAW | $-3.289$ $[-3.494, -3.081]$ | $+1.188$ $[+0.985, +1.386]$ | $2.77 \times 10^{-3}$ | 0.0 % | $1.42 \times 10^{-2}$ |
| SBS | $-4.298$ $[-4.436, -4.156]$ | $+0.178$ $[+0.097, +0.258]$ | $5.80 \times 10^{-4}$ | 1.6 % | $2.34 \times 10^{-3}$ |
| SBS-B | $-4.477$ $[-4.569, -4.364]$ | $+0.000$ $[+0.000, +0.000]$ | $2.91 \times 10^{-4}$ | 0.3 % | $1.65 \times 10^{-3}$ |
| SIZE-B | $-4.374$ $[-4.506, -4.217]$ | $+0.102$ $[+0.020, +0.192]$ | $5.31 \times 10^{-4}$ | 1.5 % | $2.33 \times 10^{-3}$ |
| ENet (D18+S) | $-4.539$ $[-4.646, -4.410]$ | $-0.062$ $[-0.104, -0.020]$ | $3.08 \times 10^{-4}$ | 1.2 % | $1.43 \times 10^{-3}$ |
| ENet (D18) | $-4.512$ $[-4.622, -4.376]$ | $-0.036$ $[-0.082, +0.012]$ | $3.66 \times 10^{-4}$ | 1.5 % | $1.62 \times 10^{-3}$ |
| ENet (size_budget) | $-4.537$ $[-4.655, -4.410]$ | $-0.061$ $[-0.123, -0.008]$ | $4.58 \times 10^{-4}$ | 0.8 % | $1.70 \times 10^{-3}$ |
| HGB (D18+S), frozen | $-4.565$ $[-4.672, -4.435]$ | $-0.088$ $[-0.145, -0.027]$ | $3.14 \times 10^{-4}$ | 0.5 % | $1.32 \times 10^{-3}$ |
| HGB (D18) | $-4.487$ $[-4.618, -4.346]$ | $-0.011$ $[-0.094, +0.068]$ | $6.61 \times 10^{-4}$ | 1.0 % | $1.81 \times 10^{-3}$ |
| HGB (size_budget) | $-4.327$ $[-4.477, -4.146]$ | $+0.150$ $[+0.046, +0.263]$ | $1.58 \times 10^{-3}$ | 1.6 % | $2.70 \times 10^{-3}$ |
| MLP (D18+S) | $-4.404$ $[-4.531, -4.245]$ | $+0.073$ $[-0.003, +0.155]$ | $2.02 \times 10^{-3}$ | 1.5 % | $2.57 \times 10^{-3}$ |
| MLP (D18) | $-4.187$ $[-4.387, -3.954]$ | $+0.290$ $[+0.123, +0.476]$ | $4.09 \times 10^{-3}$ | 3.3 % | $1.47 \times 10^{-2}$ |
| MLP (size_budget) | $-4.282$ $[-4.493, -4.034]$ | $+0.194$ $[+0.044, +0.379]$ | $2.48 \times 10^{5}$ | 3.1 % | $1.48 \times 10^{-2}$ |
| Ridge (D18+S) | $-4.542$ $[-4.641, -4.426]$ | $-0.066$ $[-0.113, -0.023]$ | $4.84 \times 10^{-4}$ | 1.4 % | $1.40 \times 10^{-3}$ |
| Ridge (D18) | $-4.494$ $[-4.602, -4.370]$ | $-0.017$ $[-0.064, +0.030]$ | $3.20 \times 10^{-4}$ | 1.7 % | $1.62 \times 10^{-3}$ |
| Ridge (size_budget) | $-4.533$ $[-4.650, -4.406]$ | $-0.056$ $[-0.117, -0.003]$ | $4.32 \times 10^{-4}$ | 0.8 % | $1.69 \times 10^{-3}$ |
| Structured (D18+S) | $-4.469$ $[-4.578, -4.347]$ | $+0.007$ $[-0.052, +0.060]$ | $5.06 \times 10^{-4}$ | 1.7 % | $1.89 \times 10^{-3}$ |
| Structured (D18) | $-4.397$ $[-4.515, -4.266]$ | $+0.080$ $[+0.031, +0.125]$ | $6.85 \times 10^{-4}$ | 2.3 % | $2.73 \times 10^{-3}$ |
| Structured (size_budget) | $-4.332$ $[-4.461, -4.185]$ | $+0.144$ $[+0.082, +0.211]$ | $1.03 \times 10^{-3}$ | 2.7 % | $4.34 \times 10^{-3}$ |
| Empirical reference | $-4.756$ $[-4.840, -4.654]$ | $-0.279$ $[-0.334, -0.228]$ | $1.98 \times 10^{-4}$ | 0.0 % | $1.13 \times 10^{-3}$ |

All budgets are averaged within a base circuit, then circuits receive equal weight. The primary objective is l = log10(MSE + $10^{-6}$); $\Delta\ell$ = l(policy) - l(SBS-B), paired percentile bootstrap over base circuits (1000 resamples). A difference $\Delta\ell$ changes the geometric mean of (MSE + $10^{-6}$) by the factor $10^{\Delta\ell}$; it is not a percentage change of the MSE. Mean MSE: arithmetic, circuit-weighted. Severe (risk > 10 x RAW's risk): declared severe degradation (share of rows); q95: 95 % quantile of the chosen risk. D18 = deployable features of 1.8.0; D = the deployable features of 1.7.0 (without the 1.8.0 noise-composition features), kept as frozen comparators; +S = with simulator-only features. Out-of-fold predictions (folds grouped by base circuit); the frozen configuration was chosen on these numbers, so they are not confirmatory.

**Table 5:** Representative per-candidate MSEs at three total shot budgets

| Candidate | $B = 10^3$ | $B = 10^5$ | $B = 10^7$ |
|---|---|---|---|
| RAW | $1.21 \times 10^{-3}$ [88] | $4.62 \times 10^{-4}$ [88] | $4.54 \times 10^{-4}$ [88] |
| RO | $1.44 \times 10^{-3}$ [88] | $1.67 \times 10^{-4}$ [88] | $1.54 \times 10^{-4}$ [88] |
| ZNE-L | $3.80 \times 10^{-3}$ [88] | $4.35 \times 10^{-5}$ [88] | $1.62 \times 10^{-6}$ [88] |
| ZNE-R | $1.16 \times 10^{-2}$ [88] | $1.18 \times 10^{-4}$ [88] | $2.03 \times 10^{-6}$ [88] |
| ZNE-E | $4.11 \times 10^{-3}$ [88] | $4.39 \times 10^{-5}$ [88] | $1.77 \times 10^{-6}$ [88] |
| PEC | $1.55 \times 10^{-3}$ [88] | $1.74 \times 10^{-5}$ [88] | $2.30 \times 10^{-6}$ [88] |
| CDR | —[0] | $5.42 \times 10^{-5}$ [82] | $2.00 \times 10^{-6}$ [82] |
| NIL | —[0] | $3.19 \times 10^{-3}$ [70] | $4.12 \times 10^{-3}$ [70] |
| RO-TREX | $1.11 \times 10^{-3}$ [88] | $1.61 \times 10^{-4}$ [88] | $1.50 \times 10^{-4}$ [88] |
| RO-TW | $1.44 \times 10^{-3}$ [88] | $1.73 \times 10^{-4}$ [88] | $1.68 \times 10^{-4}$ [88] |
| RO-DD | $1.41 \times 10^{-3}$ [88] | $1.74 \times 10^{-4}$ [88] | $1.68 \times 10^{-4}$ [88] |
| ZNE-L-TW | $4.05 \times 10^{-3}$ [88] | $4.31 \times 10^{-5}$ [88] | $1.27 \times 10^{-6}$ [88] |
| ZNE-E-B | $4.11 \times 10^{-3}$ [88] | $4.13 \times 10^{-5}$ [88] | $1.08 \times 10^{-6}$ [88] |
| PEC-TW | $1.35 \times 10^{-3}$ [88] | $1.69 \times 10^{-5}$ [88] | $1.77 \times 10^{-6}$ [88] |
| FR | —[0] | $1.03 \times 10^{-2}$ [81] | $9.61 \times 10^{-3}$ [81] |
| SV | $6.42 \times 10^{-4}$ [16] | $1.33 \times 10^{-5}$ [16] | $3.38 \times 10^{-6}$ [16] |
| ZNE-L$\circ$SV | $2.29 \times 10^{-3}$ [16] | $2.04 \times 10^{-5}$ [16] | $2.99 \times 10^{-7}$ [16] |

Median evaluation-half MSE over the feasible development tasks, with the number of feasible tasks in brackets (out of 88). Candidates have different feasible subsets: this table is descriptive and must not be read as a ranking; paired comparisons are in the confirmation and development tables.

**Table 6:** Infeasibility, fallbacks, severe degradation and loss concentration per candidate

| Candidate | Infeasible/error | Flagged trials | No estimate | $R > 10R_{\mathrm{RAW}}$ | $q_{95}$ MSE | Worst 5 % |
|---|---|---|---|---|---|---|
| RAW | 0.0 % | 0.0 % | 0.0 % | 0.0 % | $1.42 \times 10^{-2}$ | 43 % |
| RO | 0.0 % | 0.0 % | 0.0 % | 1.0 % | $1.03 \times 10^{-2}$ | 51 % |
| ZNE-L | 0.0 % | 0.0 % | 0.0 % | 0.0 % | $3.78 \times 10^{-3}$ | 17 % |
| ZNE-R | 0.0 % | 0.0 % | 0.0 % | 11.4 % | $1.16 \times 10^{-2}$ | 10 % |
| ZNE-E | 0.0 % | 6.3 % | 0.0 % | 2.4 % | $4.44 \times 10^{-3}$ | 89 % |
| PEC | 0.0 % | 0.0 % | 0.0 % | 0.9 % | $1.70 \times 10^{-3}$ | 26 % |
| CDR | 16.1 % | 4.4 % | 0.0 % | 2.6 % | $2.76 \times 10^{-3}$ | 32 % |
| NIL | 44.3 % | 8.8 % | 0.0 % | 47.9 % | $1.72 \times 10^{-1}$ | 39 % |
| RO-TREX | 0.0 % | 0.0 % | 0.0 % | 0.9 % | $1.03 \times 10^{-2}$ | 52 % |
| RO-TW | 0.0 % | 0.0 % | 0.0 % | 1.4 % | $1.02 \times 10^{-2}$ | 50 % |
| RO-DD | 0.0 % | 0.0 % | 0.0 % | 1.0 % | $1.11 \times 10^{-2}$ | 49 % |
| ZNE-L-TW | 0.0 % | 0.0 % | 0.0 % | 0.0 % | $4.02 \times 10^{-3}$ | 17 % |
| ZNE-E-B | 0.0 % | 6.3 % | 0.0 % | 2.4 % | $4.44 \times 10^{-3}$ | 23 % |
| PEC-TW | 0.0 % | 0.0 % | 0.0 % | 0.1 % | $1.66 \times 10^{-3}$ | 28 % |
| FR | 17.2 % | 3.4 % | 0.0 % | 61.0 % | $2.78 \times 10^{1}$ | 83 % |
| SV | 81.8 % | 0.0 % | 0.0 % | 2.5 % | $5.50 \times 10^{-4}$ | 25 % |
| ZNE-L∘SV | 81.8 % | 0.0 % | 0.0 % | 0.0 % | $1.92 \times 10^{-3}$ | 20 % |

Development set, all budgets. Infeasible/error: share of (task, budget) rows without a feasible allocation or with a recorded method error. Flagged trials: share of the evaluation trials of feasible rows with any method fallback or failure flag (part of the risk, never removed); No estimate: nonresponse share. Severe (risk > 10 x RAW's risk): circuit-weighted share of feasible rows whose risk exceeds ten times RAW's on the same row (declared before the 1.8.0 results). q95: 95 % quantile of the risk over feasible rows. Worst 5 %: share of the candidate's total circuit-averaged loss carried by its worst 5 % of base circuits. Each candidate on its own feasible subset (descriptive).

**Table 7:** Classical cost of one recommendation (seconds, median of repeats)

| $n$ | Compile | Features | Struct. pref. | Design pref. | Inference | Whole call | Sim. features | Labels (study) |
|---|---|---|---|---|---|---|---|---|
| 3 | 0.0088 | 0.0139 | 0.000335 | 0.0669 | 0.0261 | 0.0394 | 0.225 | 38.2 |
| 4 | 0.00915 | 0.0189 | 0.000474 | 0.114 | 0.0297 | 0.0483 | 0.396 | 35.1 |
| 5 | 0.00936 | 0.022 | 0.000443 | 0.137 | 0.0271 | 0.0486 | 0.442 | 38.6 |
| 6 | 0.0102 | 0.0324 | 0.000612 | 0.189 | 0.0263 | 0.0564 | 0.634 | 59.4 |
| 7 | 0.0105 | 0.0358 | 0.000542 | 0.324 | 0.0272 | 0.0591 | 0.902 | — |

Measured on the build machine (2 shared cores): magnitudes, not a benchmark. ising circuit, fake_fez, B = $10^5$. Features: the deployable features without cache; Struct./Design pref.: the structural preflight of every candidate and the design-level preflight of the learning candidates; Whole call: deploy.recommend with the default bundle; Sim. features: the simulator-only features (exponential in n; only for a model that uses them); Labels: generating one development task's benchmark labels (a study cost, not a recommendation cost). No component uses quantum shots. Early rejection (whole call, GHZ-type chain outside the training widths): n = 12: 2.02 s, out-of-distribution; n = 24: 4.64 s, out-of-distribution; n = 40: 16.49 s, out-of-distribution

**Table 8:** Original against expanded portfolio on identical development tasks

| Contrast | $\Delta\ell$ [95%] | Factor $10^{\Delta\ell}$ |
|---|---|---|
| More candidates, fixed policy (SBS-B 1.8 - SBS-B 1.6) | $-0.070$ $[-0.133, -0.012]$ | 0.851 |
| More candidates, attainable (reference 1.8 - reference 1.6) | $-0.092$ $[-0.165, -0.037]$ | 0.809 |
| Learning within portfolio-1.6 (Ridge - SBS-B) | $+0.015$ $[-0.033, +0.066]$ | 1.035 |
| Learning within portfolio-1.6 (HGB - SBS-B) | $+0.002$ $[-0.057, +0.063]$ | 1.005 |
| Learning within portfolio-1.8 (Ridge - SBS-B) | $-0.017$ $[-0.064, +0.030]$ | 0.962 |
| Learning within portfolio-1.8 (HGB - SBS-B) | $-0.011$ $[-0.094, +0.068]$ | 0.976 |
| Total (Ridge 1.8 - SBS-B 1.6) | $-0.087$ $[-0.175, -0.009]$ | 0.818 |
| Total (HGB 1.8 - SBS-B 1.6) | $-0.081$ $[-0.197, +0.019]$ | 0.831 |

Out-of-fold development results (880 rows, 44 base circuits; feature set deployable18; every policy refitted per portfolio). $\Delta\ell$: paired difference of the mean over base circuits of log10(MSE + $10^{-6}$), percentile bootstrap over base circuits; the factor is the change of the geometric mean of (MSE + $10^{-6}$). Exploratory.